Department of Computer Science

National University of Computer and Emerging Sciences

Lahore Campus

Data Mining

Section A

Quiz 3a

Solution

Question 1: 4 marks

Use the following data to calculate Error rate, precision, recall, and specificity.

A	**PREDICTED CLASS**		
ACTUAL CLASS		**Class=Yes**	**Class=No**
	Class=Yes	**40**	**10**
	Class=No	**10**	**40**

Precision $(\mathrm{p}) = 0.8$TPR $=$ Recall $(\mathrm{r}) = 0.8$FPR $= 0.2$F−measure $(\mathrm{F}) = 0.8$Accuracy $= 0.8$
TPR/FPR = 4

Question 2: 3 marks

Input

1	0	1	0	2
1	1	3	2	1
1	1	0	1	1
2	3	2	1	3
0	2	0	1	0

1	0	0	1	0
2	0	1	2	0
3	1	1	3	0
0	3	0	3	2
1	0	3	2	1

2	0	1	2	1
3	3	1	3	2
2	1	1	1	0
3	1	3	2	0
1	1	2	1	1

Kernel

0	1	0
0	0	2
0	1	0

2	1	0
0	0	0
0	3	0

1	0	0
1	0	0
0	0	2

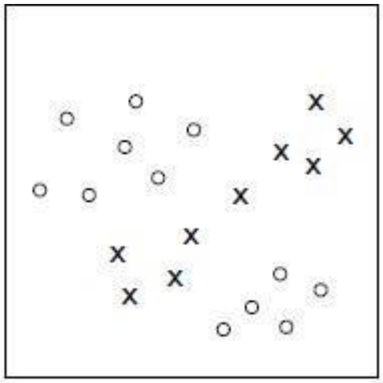
Perform convolution using the above matrices for a 3-channel RGB input, stride = 1, no padding and bias as 5.

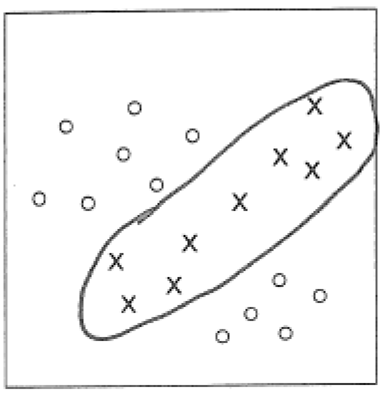
https://medium.com/apache-mxnet/multi-channel-convolutions-explained-with-ms-excel-9bbf8eb77108

Make sure that you consider the bias which is 5.

Question 3: 3 marks

For each of the following example of training data, (1) sketch a classification boundary that separates the data; (2) state whether or not the data are linearly separable, and if not, (3) give a set of features that would allow the data to be separated. (Your features do not need to be minimal, but should not contain any obviously unneeded features.)





No - not linearly separable

Could use features such as

$$[1 \quad x_1 \quad x_2 \quad x_1x_2 \quad x_1^2 \quad x_2^2]$$